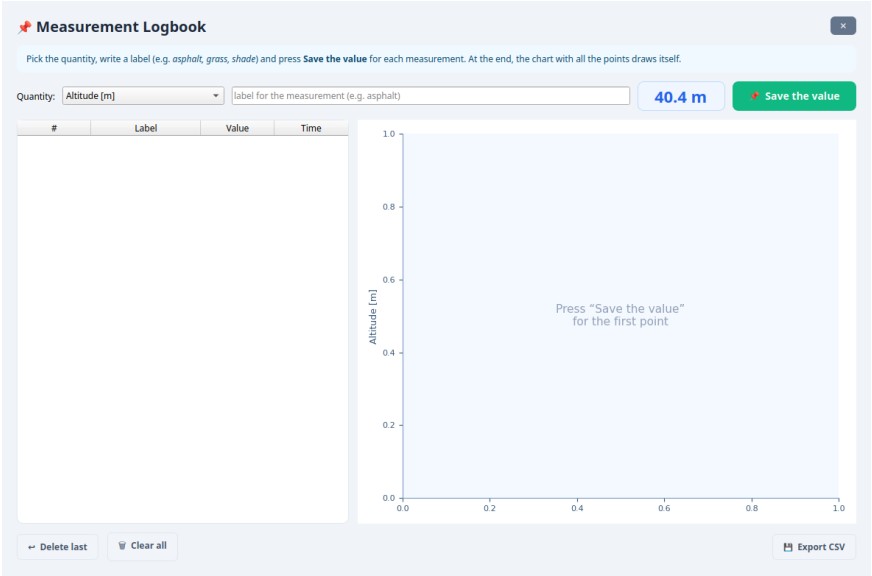
Measurement Notebook

A label, a place, a time and the conditions — that is all that separates a number from data

Level: Explorer and Scientist — ages 10–18 (Romanian school years IV–XII) • **Time in class:** 20 min, one 50-minute lesson or a week-long project • **Sensors:** All sensors: BME280, light sensor, SCD41, MPU-6500



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 10–18 (Romanian school years IV–XII). The same app serves every age, but the task differs: at **support**, students receive the route, the label list and the quantity ready-made; at **core**, students propose their own labels for a given question; at **extension**, they measure two quantities at every point and look for a link between them. **This is not a content lesson but a method lesson**: the most direct curriculum entry is at **year VI**, physics, where measuring physical quantities — instrument, unit, repetition — is explicit syllabus material.

Assumed knowledge

Reading a number with a unit and writing it into a table. For the full route, in addition: the arithmetic mean and the range. For the project: working in a spreadsheet. For support: only copying the number from the screen into the already-labelled row.

Time in class

Demonstration: 20 min. Full lesson: 50 min. Project: a school week — one 50-minute lesson at the start, three days of 15 minutes of routine and one 50-minute lesson of analysis, that is 145 minutes in total (2 h 25 min).

Teacher preparation

30 min — **walking the route yourself first**, sticking numbered labels at every stop, deciding the measuring height, printing the room plan and the **exact label list**, creating the shared folder and fixing the file-naming rule.

Sensor settling time	**60 seconds at every stop** for temperature and humidity, **90 seconds** for CO_2, and practically **nothing** for illuminance. This rule alone is the difference between a data set and a heap of numbers.
Sensors and boxes	Any box: the app lists exactly the quantities the box sends. **The same box for the whole of a series** — two units can differ by a few tenths of a degree. One box per group; the CO_2 protocol needs the SCD41.

Three routes

Short	20 min: a demonstration in a single room — six labelled points, the chart viewed together, the export opened in front of the class.
Full	50 min: one of the three protocols, taken to completion — written prediction, the circuit walked twice, export, first analysis.
Project	A week: the same protocol, the same route, the same time, five days, one dated file per day and one lesson of analysis at the end.

1. Where this fits in the curriculum

The Measurement Notebook is the most general app in the collection, and it does one thing, very well: it captures a single reading the instant a student presses a button, attaches the label the student has written, and stamps it with the time. It draws a chart of every point as it goes and exports them to an Excel (`.xlsx`) or CSV file. It is the digital equivalent of a clipboard and a pencil, but with a calibrated instrument and no transcription errors.

- **Physics** (*Fizică*) — at **year VI**, where **measuring physical quantities** (instrument, unit, repeating the measurement, approximating the result) is explicit syllabus material. This is the most direct entry, and the only one that is not an application: the activity *is* the lesson about how a measurement is made.
- **Geography** (*Geografie*) — at **year V**, observation and measurement in the local horizon; at **years IX–XII**, the geographic field method: sampling, the map as a data support, microclimate. The heat map of the yard or the classroom is a direct application of these.
- **Mathematics** (*Matematică*) — the arithmetic mean and the range at **years V–VI**; **representing data** in a table and a bar chart at **years VI–VII**; and at **year VII**, points on a coordinate system, that is, the scatter plot from the extension route.
- **Computer science and ICT** (*Informatică și TIC*) — **years V–XII**, processing data in a spreadsheet: sorting by label, computed columns, means, comparison charts across two days. The exported file opens directly.
- **Cross-curricular** — health education (air quality and light at the place of study), technology education (auditing a building and a reasoned proposal to the school's leadership), and the habit of recording conditions alongside a number.

The core skill is **disciplined sampling**. Any student can read a display. What is learned here is that a reading is worth nothing unless you can say where it was taken, when and under what conditions — and that **the label a student writes is just as much a part of the data as the number is**.

2. Time plan

Three modes, from a single demonstration to a week-long project. Choose *one* column and stay in it; mixing them halfway through loses exactly the comparability that makes the data interesting.

Route	Time	What happens
Short	**20 min**	A demonstration in a single room, with no walking. **Temperature** is chosen from the drop-down list and six labelled readings are taken: windowsill, radiator, room centre, floor, head height, schoolbag (4). The chart of the points is viewed together (6). The file is exported and opened in front of the class (5). Discussion of what makes a good label (5).
Full	**50 min**	One of the three protocols, taken to completion: the written prediction and discussing it (7), the briefing — the route, the label list, the 60 s wait (5), the circuit walked **twice** (25), the export and the first analysis (13).
Project, day 1	**50 min**	The full route above, plus fixing the file-naming rule and appointing, on rotation, the **data officer**. Day 1's file is the reference.
Project, days 2–4	3×15 min	The duty group walks the same route, at the *same time*, with the same labels in the same order, exports and puts the dated file in the shared folder.
Project, day 5	**50 min**	All the files into one spreadsheet, every place tracked across days, normalising to the day's mean, and answering the question: *which places are consistently different, and which merely vary?*

The project means **two 50-minute lessons** (the first and last day) and **three days of 15 minutes** of routine — that is $50+50+3\times15 = 145$ minutes, **two hours and 25 minutes** spread over a school week. It can stretch to two weeks by adding five more days of routine, but five days is the minimum that says anything.

Walking is the expensive part — walk the route yourself first

Six stops spread through the building mean about 15 minutes of clean walking, before any waiting at all. Walk the route yourself, once, before the lesson; time it; stick on the numbered labels; and give every group the **printed order of the stops**. A group that has to decide for itself where to go next collects half the data.

The practical rule for choosing a protocol: if the route leaves the room, allow 60 seconds of waiting *plus* one minute of walking for every stop. Within a single room, the walking almost disappears, and that is why the classroom heat-map protocol is the safest bet for a first lesson.

3. Materials

What you need	Per group	Notes
MagicBox	1	The app lists exactly the quantities the box sends, so a full box gives the widest choice. **The same box for the whole of a series.**
Laptop with the app	1	The box must stay connected while readings are captured, so the laptop travels with the group.
Clipboard and the **printed sheet**	1	The app remembers only the label, the value and the time. **The place, the height and the conditions are written by hand** — there is nowhere else for them.
The label list, printed and taped to the clipboard	1	The object that saves the project. Without it, half the class writes `window` and the other half `sill`, and the export can no longer be grouped.
Numbered sticky labels	8–10	Stuck at every measuring point, so that every group and every day measures in exactly the same place.
Room or floor plan	1	Printed A4. Points are numbered on the plan, and those numbers are the ones found in the labels.
Tape measure	1	For the height above the floor and the distance from the window or radiator. Reproducibility depends on it.
Stopwatch (or a student's phone)	1	The 60 seconds are timed, not estimated.
Ordinary thermometer	1 per class	Cross-check. Comparing with a familiar instrument quickly settles questions of trust.
Power bank and a shared folder	1 per class	Power for routes without sockets; a single folder that all the dated files go into.

Preparation: walk your route and fix the measuring points *before* the first lesson, sticking a numbered label at each one. Decide the height above the floor — 1.1 m, the head height of a seated student, is a good choice — and write it on the label. Switch the box on ten minutes before the lesson, so the sensors warm up. Make the shared folder — for example `logbook-heat-map/` — and **fix the file-naming rule in the first session, not the last**. And print the label list: it is the one object in the table above without which the project cannot be saved at the end.

4. Scientific background for the teacher

Why a single labelled reading is a genuine measurement

A single, instantaneous reading carries no information. It becomes evidence only once three things travel with it: **where**, **when** and **under what conditions**. The app stamps the time and the value from a calibrated sensor on its own; the student supplies the place, through the label. This is exactly the structure of a scientific data set: a dependent quantity, a *categorical* independent variable, and a set of conditions noted separately.

When two students compare two labelled points, they are making an unpaired comparison. The honest question is always whether the difference between the two numbers is bigger

than the variation you would get from measuring the *same* place twice. That is why the first thing taught is **repetition**: the same point measured twice, a few minutes apart. If readings at the same place differ by ±0.3 °C, then a difference of 0.2 °C between two places means nothing, and one of 4 °C means a great deal.

4.1 Response time is not zero

Every sensor has a time constant — how long it takes to reach equilibrium with its new surroundings after being moved. Roughly:

- **Illuminance (light sensor)** — practically instantaneous, a second or two at most (with the new firmware the sensor picks its own sensitivity). Move it and read it.
- **Temperature and humidity (BME280)** — tens of seconds up to a minute or two for a large change, because the whole board and casing must change temperature, not just the chip.
- **Pressure (BME280)** — fast, but the quantity itself is extremely sensitive: opening a door in a ventilated building moves it.
- **CO_2 (SCD41)** — **a new reading every five seconds** and about a minute to settle after a large change in concentration.

The practical rule for class: **stand still and wait** 60 **seconds before pressing Save the value**, for anything except light, and 90 **seconds for CO_2**. A box brought in from a cold corridor into a warm room and read at once reports the corridor, not the room. This one rule alone is the difference between a data set and a heap of numbers.

4.2 What “the same place” means

Microclimates are small. The air temperature 20 cm above a radiator and 20 cm to its side can differ by several degrees; illuminance halves within a metre of a window. Reproducibility therefore needs three things written down: **the label stuck on the wall**, **the declared height** and **the declared orientation**. If the box is held in a warm hand, pressed to a chest, it measures the student.

4.3 How to read the chart of points

The chart the app draws puts one *point* for each reading: *labels* sit on the horizontal axis, not a continuous quantity, and the points appear **in the order they were captured**. A chart ordered from coldest to warmest therefore needs either a route planned in advance, or sorting the exported file.

The vertical axis does not start at zero. It zooms in on the range of the data, precisely so that a small difference can be seen: 41.5 and 41.9 appear as two clearly separate points, not as two nearly equal bars. That is the right choice for *seeing* differences, but it has a price students need to know: on a zoomed axis, a difference of two tenths of a degree looks as impressive as one of four degrees. **Always read the numbers on the axis**, not the height of the drawing, and judge the difference against the threshold from the investigation (spread plus drift). The order of the points is also a presentation choice, which can make a difference look bigger or smaller than it is.

Two identical labels give **two separate points**: the app does not add or average anything. This is good news — repetition stays visible — and a trap: if someone writes the same label twice for two different places, the two can never be told apart again.

4.4 Why the label list is the most important practical part

In a spreadsheet, `window`, `Window`, `windo` and `by-window` are **four categories**, and inside a student's head they are one. A class in which half write `sill` and the other half `window` can no longer group the export, and all the fieldwork is lost at the analysis stage.

This is why this guide does not say "agree on some labels", but gives the **exact, printable list**, for each of the three protocols, together with six writing rules. The rule about plain, unaccented letters is worth explaining to students: it has nothing to do with correct spelling, and everything to do with the fact that a stray capital letter, an extra space or a typed apostrophe turns a matching label into a new, unmatched category, silently and without any warning. *On the sheet we write normally; in the label field we write exactly as printed, on purpose.*

5. Lesson flow

The Measurement Notebook is not a step-by-step wizard. There is no green button that advances it, no **Back**, no **Finish**, no observation log — it is a window that runs for as long as you leave it open. The pace is set by you, through the protocol.

Technical note — the window and the buttons, before the first reading

How to start it. From the **Activities** page, with the **Start** button on the **Measurement Notebook** card. The app requires a connected box; if none is connected, it refuses with a message asking you to connect a box before launching the activity. If more than one box is connected, a picker appears first, with the boxes under the names given to them in **Settings**. The window that opens carries the same title as the activity, followed by the box's name; with several boxes connected, the **Change box:** list in the header moves the notebook to another box. Below the header, a blue strip repeats briefly what to do.
The Quantity: drop-down list, top left. **It is empty for the first few seconds** and fills only once data arrives from the box — wait for it. It then contains exactly the quantities the box sends, **with the most used ones first**. For a full box, the order is: **Temperature [°C]**, **Humidity [%]**, **Pressure [hPa]**, **Altitude [m]**, **Illuminance [lx]**, **CO_2 [ppm]**, **Temperature (SCD41) [°C]**, **Humidity (SCD41) [%]**, then the three accelerations and the three gyroscopes. **Temperature comes first and is chosen from the start.**
The label field, in the middle, with the placeholder text **label for the measurement (e.g. asphalt)**. **The Enter key saves the point**, exactly like the button — type the label, press Enter, the field clears itself for the next one. It is the fast method and is worth demonstrating.
The big blue number, to the right of the field, is the **live value**, with its unit. **It is not yet saved.** While the box sends nothing, a dash appears there instead.
The green Save the value button captures the value at the instant it is pressed. The point appears immediately in the table on the left, with the columns **#**, **Label**, **Quantity**, **Value** (with its unit) and **Time** (with seconds), and as a new point in the chart on the right.
Every point remembers its quantity: the **Quantity** column says what was measured, however the list is changed later.
The chart on the right shows only the points of the quantity chosen in the list, each with its value written above it and its label below. The **Show differences from the first point** box, above it, turns it into differences from the first point, with a line at zero — the clearest way to see a small difference.
Below, three buttons: Delete last, **Clear all** and **Export data**. The window closes with the **Close** button in the top right corner.
What does not exist here. There are no steps, no **Back**, no **Finish**, and none of the live charts' **Smooth**, **Pause** or **Clear** buttons. Those are the buttons of the guided activities and of the live charts; do not look for them here.

5.1 The short route: the 20-minute demonstration

Project the screen and stay in the room. Choose **Temperature** from the drop-down list and take six readings, with 60 seconds of waiting at each, with the labels written on the board before you start: `windowsill`, `radiator`, `centre`, `floor`, `head-height`, `schoolbag`.

Ask the class, *beforehand*, to predict the order from warmest to coldest. Then look at the chart of the points together and compare it with the prediction — reading the numbers on the axis, because the axis does not start at zero. Ask one hard question: *how big does a difference between two points have to be before it deserves to be believed?* — and leave it open, because its answer is exactly the protocol for the next lesson. Close by pressing **Export data** and opening the file in front of the class: a file opened once, in front of everyone, makes the rest of the project credible.

Technical note — right at the step where someone wants to change the quantity

Changing the quantity is safe. Every point keeps the quantity it was captured with: in the table, in the **Quantity** column, and on export, in the `quantity` column. Changing the **Quantity:** list relabels nothing; it only changes what the chart shows — **only the points of the chosen quantity**. If the chart looks empty after a change, the other points are not lost: they are in the table and come back when their quantity is chosen again.
The working rule is still a single one: one quantity per investigation. Not because the app would mix up the data, but because a map of the room compares point with point only if every point is the same quantity, captured with the same wait. When you measure two quantities at the same place (the extension route), capture them one after the other, with the same label.
The two delete buttons are not the same. Delete last removes only the last row, without asking — the button for a wrong label spotted immediately. **Clear all** asks first ("Delete all … points? Export them first if you need them."), but once confirmed it **cannot be undone**. Name a single person in the group who is allowed to touch it.
Export before closing the window. From v5.3 the app also keeps every saved point in a file in the `logs` folder, but that is a safety net, not the project file: **export before closing**, not after, with the name following the class's rule.

Technical note — the export, the filename and the five columns

The button. Export data, bottom right. It opens a save dialogue with a suggested name built from the box's identifier, the date and the time — for example `logbook_BOX_3859_20260914_0910.xlsx`. **The default format is Excel (`.xlsx`)**, which opens correctly in any Excel; `.csv` can also be chosen, and when saved while the app is in Romanian, French or German it uses ";" and a decimal comma. If no point was captured, the message **There are no measurements to export** appears; on success, **Saved:** followed by the file's path. If the chosen file is open in Excel, saving fails and the app says why — close it and try again.
The file has five columns, in this order: `label` (the label the student wrote), `quantity` (**the quantity's internal code** — `T`, `H`, `P`, `LUX`, `CO2`), `value` (the value, with every decimal the box sent), `unit` (the unit) and `time` (the date and time, with seconds). The `quantity` column holds **the code, not the displayed name**, so write the correspondence on the board before the first analysis: `T` is temperature, `LUX` is illuminance, `CO2` is carbon dioxide. A file with two quantities is split simply by filtering on this column.
Rename on saving, following a rule the class decided in the first session, with **the date first**, in `YYYY-MM-DD` form, so that sorting alphabetically also sorts chronologically:

`heat-map_2026-09-14_0910_groupA.csv`

All the project's files go into a single folder, not one folder per day.
The safety net, and its limits. The app writes, on its own, into the `logs` folder, the

box's raw stream and, from v5.3, every saved point, the moment it is saved, in a file `notebook_<box>_<date>.csv` (columns `time`, `label`, `quantity`, `value`, `unit`; a plain CSV, with commas between columns and a decimal point, whatever the app's language). Points removed with **Delete last** or **Clear all** stay in that file. Files there are no longer deleted automatically (automatic deletion is off by default, in **Settings**). It is a genuine safety net for when a group forgets to export, but it is never the plan: the filenames do not follow the class's rule and they are not in the shared folder.

5.2 The week-long project

Three rules make the difference between a data set and a heap of numbers.

The same time every day. Temperature, light and CO_2 follow strong daily cycles. A reading at 09:00 on Monday and one at 14:00 on Tuesday says something about the time of day, not about Monday and Tuesday.

The same order of stops. The circuit itself takes 10–15 minutes in a single room and up to 20 if it leaves it, during which the conditions change. If the order changes, the drift becomes noise that can no longer be separated from the effect.

The same labels, written identically, and the same box. The printed list stays taped to the clipboard. And every day, one line is written down about the weather and the state of the room: sun or cloud, windows open or closed, heating on or off, how many people. *Half of the differences between days will be explained by that line.*

Gaps — a weekend, a day the box would not connect — are not a problem, they are data. The Monday-morning reading, after two days with no heating and no people, is one of the most informative points in the whole series. Mark it in the spreadsheet with a note; do not discard it.

For handovers between groups, keep a laminated **protocol sheet** beside the box: the route, the label list, the time, the height and the naming rule. Any group should be able to take over from that sheet alone. Name, in every group, a **data officer**, who is responsible for the export and for putting the file in the folder, and rotate the role.

The investigation, in one table

The question. Does our room have the same temperature everywhere, or not?
The hypothesis. Written on the sheet **before** the first reading: the warmest place will be _____, the coldest _____, and the difference between them will be _____ °C. It is not changed afterwards.
What I change. The place, and nothing else. This is a *categorical* variable: it does not have big and small values, it has names.
What I measure. Air temperature, in °C, at each of the eight stops, **twice** — once on the first circuit and once on the second.
What I keep the same. The same height above the floor, measured with the tape measure, not guessed; **the same box**; **the same order** of stops; **the same 60-second wait**, timed; the box held by its edges, away from the body, in the shade of the window; and the windows and door in the same state throughout the circuit.
How many repeats. Two readings at every stop, obtained by walking the whole circuit twice — **not** by pressing the button twice in the same place, which measures nothing new.
What makes a result valid — and what makes it an artefact. You have two numbers to calculate before declaring anything: **the repeat spread** (how much the two readings from the same place differ) and **the drift** (how much the first stop changed between the first and second circuit — that is, how much the whole room warmed up while you were walking).

A difference between two places counts only if it is bigger than the spread and the drift added together. If it is not, the correct conclusion is *"we could not detect a difference with our setup"*, not *"there is no difference"*. And if every stop rose by the same amount, that is not the room's map: it is the time that passed.

6. The three ready-to-use protocols

Each protocol fits into a 50-minute lesson, uses the app exactly as it is, and comes with **its own exact label list**. Choose *one* and carry it through to the end.

The six label-writing rules, the same across all three — print them and tape them to the clipboard:

1. only **lower-case letters**;
2. **no accented letters** (`naive`, not `naïve`);
3. words joined with a **hyphen**, never a space;
4. at most **three words**;
5. only **from the printed list** — nothing invented on the spot;
6. **never leave the field empty**: the app writes `#1`, `#2`, `#3` on its own, and those can never be grouped and can no longer be recognised the next day.

6.1 Protocol 1: the classroom heat map

The question. Does our room have the same temperature everywhere?

The quantity in the list. Temperature [°C].

The route. Eight stops, in this order, at 1.1 m above the floor, measured with the tape measure. The circuit is walked **twice**, in the same order. Wait: 60 **seconds** at every stop.

The exact label list — eight, in the order walked:

#	Label, exactly this	Where exactly
1	`door`	By the door, one step inside. **First and second circuit both** — the difference between the two readings is the drift.
2	`front-left-corner`	The corner on the left, near the board.
3	`teacher-desk`	Beside the teacher's desk, not on it.
4	`front-right-corner`	The corner on the right, near the board.
5	`centre`	The middle of the room.
6	`window`	One step from the window, not on the sill.
7	`radiator`	20 cm to the *side* of the radiator, never above it.
8	`back-corner`	The corner at the back of the room, furthest from the window.

What is written on paper (the app keeps none of this): the height measured, whether the heating is on, whether the windows are open, how many people are in the room and the time the circuit started.

Expected result. Differences of 1–4 °C within a single room are normal; near a sunlit window or a radiator that is on, more. The two readings from the same place should match within a few

tenths of a degree — ±0.3 °C is typical —, and that is exactly what gives you the right to compare two places at all. Drift over a whole circuit is usually 0.2–0.8 °C, with the class in the room.

6.2 Protocol 2: the light map

The question. Is every desk in the room lit well enough to write at?

The quantity in the list. Illuminance [lx].

Check the box's firmware first. With firmware 2.2.5 or newer, the light sensor changes its sensitivity by itself and measures up to about 120,000 lx. Older firmware saturates at around 3800 lx: if the desk by the sunny window reads a "stuck" 3800 lx or so, the box needs reflashing (see the flashing guide in the `firmware` folder), otherwise the light map cuts off exactly its bright end.

The route. A **three by three** grid: rows of desks 1, 3 and 5, and on each row the place nearest the window, the middle one and the one nearest the door. The sensor lies **flat, face up, on the desk**, never shaded by a hand or a head. **Illuminance is instantaneous**: do not wait 60 seconds here, only 5; say so explicitly, or students will wait for nothing.

The grid is walked **twice, under two conditions**: once with daylight only (lamps off, blinds up) and once with artificial light only (lamps on, blinds down). **The labels stay identical**; the condition goes *in the file name*: `light_2026-09-14_daylight_groupA.csv` and `light_2026-09-14_lamps_groupA.csv`. This is exactly the lesson about labels, learned by doing.

The exact label list — nine, in the order walked:

#	Label, exactly this	Where exactly
1	`desk-1-window`	Row 1 (nearest the board), the place towards the window.
2	`desk-1-middle`	Row 1, the middle place.
3	`desk-1-door`	Row 1, the place towards the door.
4	`desk-3-window`	Row 3, towards the window.
5	`desk-3-middle`	Row 3, middle.
6	`desk-3-door`	Row 3, towards the door.
7	`desk-5-window`	Row 5 (at the back), towards the window.
8	`desk-5-middle`	Row 5, middle.
9	`desk-5-door`	Row 5, towards the door.

What is written on paper: the time, whether it is sunny or cloudy outside, which lamps are on and whether any desk is shaded by something else.

Expected result. The usual recommendation for writing is 300–500 **lx** at desk level. Students typically find the row by the window at several thousand lux, while the furthest corner sits between 100 and 300 lx even with the lamps on — **a ratio of at least ten to one within a single room**, even though the room "looks evenly lit", because the eye adapts and the sensor does not. It is a result you can take to the school's leadership.

6.3 Protocol 3: where the air is most stagnant (needs the SCD41)

The question. Which spaces in the school have the most stagnant air?

The quantity in the list. CO_2 [ppm]. Here CO_2 is a **ventilation indicator** — how much exhaled air has built up in a space — not a dangerous substance at the levels found in a school.

The route. Seven stops, in this order, with a 90**-second** wait at each — the SCD41 gives a new reading every five seconds, so 90 seconds means eighteen readings, enough for it to settle. Everyone stays an arm's length from the box while waiting: **exhaled air is about** 40,000 **ppm** and ruins the measurement on the spot.

The exact label list — seven, in the order walked:

#	Label, exactly this	Where exactly
1	`outside`	In the yard, in the shade, away from doors and cars. **This is the reference** against which everything else is judged.
2	`empty-room`	An empty room, before lessons or right after airing it.
3	`occupied-room`	A room with a lesson under way, at its midpoint — with the agreement of the colleague teaching there.
4	`corridor-break`	The corridor, at break time, at its busiest.
5	`library`	The library or reading room.
6	`sports-hall`	The sports hall or its changing room.
7	`outside-return`	**The same place as stop 1**, at the end of the circuit. The label is deliberately different, so it appears as a separate point: the difference from `outside` is the measurement's drift.

What is written on paper: how many people were in each space, whether the windows were open and the time of each stop.

Expected result. Outside, around 420 ppm; an empty room, close to the same value; an occupied room, 800–1500 ppm and rising; a busy corridor at break, surprisingly high, but briefly. **The outdoor value is the reference** against which any indoor reading is judged — without it, 1200 ppm means nothing. The topic is treated at length in the *Air Quality* guide.

7. Analysis after the activity

7.1 What the data should look like

Place / quantity	Typical range	Notes
Air temperature in the room	19–24 °C	Spread of 1–4 °C within the same room.
Repetition at the same place	±0.3 °C	The threshold below which a difference means nothing.
Drift over a whole circuit	0.2–0.8 °C	With the class in the room; more if the heating starts up midway.
Illuminance at a desk, the furthest corner	100–300 lx	Often below recommendation.
Illuminance at a desk, the row by the window	1000–10,000 lx	Ten to a hundred times the furthest corner. With old firmware, capped at about 3800 lx.
Recommendation for writing	300–500 lx	The benchmark against which the light map is judged.
Outside, overcast sky	5000–20,000 lx	Full sun exceeds 80,000 lx; measurable only with the new firmware (up to about 120,000 lx).
CO_2 outside	around 420 ppm	The reference for any indoor reading.
CO_2, occupied room	800–1500 ppm	Rising throughout the lesson; in an unventilated room it can pass 2000 ppm — a sign of insufficient ventilation (the European standard EN 16798-1 gives the ventilation categories).
Exhaled air	about 40,000 ppm	This is why nobody breathes towards the box.

7.2 From two circuits to a conclusion, step by step

1. Open the exported file. Put the labels in one column, the values from circuit 1 in the next and those from circuit 2 in the third.
2. **Repeat spread:** for each place, calculate |circuit 2 − circuit 1|. The largest of these values is your setup's spread. Worked example: if the largest difference between the two circuits at the same place is 0.3 °C, then 0.3 is the threshold.
3. **Drift:** take the `door` place and subtract the circuit-1 reading from the circuit-2 reading. Example: $21.9 - 21.4 = +\mathbf{0.5}\,\mathbf{°C}$ — that is how much the whole room warmed up while you were walking.
4. **Mean per place:** for every label, the mean of the two circuits. This is the map.
5. **The decision rule, written on the board:** a difference between two places counts only if it is bigger than $0.3 + 0.5 = \mathbf{0.8}\,\mathbf{°C}$. Below that value, write *"we could not detect a difference"*, not *"they are the same"*.
6. **Extension:** subtract the day's mean from every reading — **normalising**. A single formula in the spreadsheet, and suddenly a warm day can be compared with a cold one, because what remains is only the map's shape.

7.3 Comparing two days

Open two dated files side by side. Put the labels in one column, the values from day 1 in the second, those from day 2 in the third and the difference in the fourth. Plot the two days as paired bars.

The rule of interpretation taught here: a place is **genuinely different** only if the change from one day to the next *at that place* is bigger than the change seen at *all* the other places. If every point rose by 2 °C, that is the weather. If one point rose by 6 °C while the rest rose by 2, that point has a story.

7.4 Questions to ask after the activity

1. Two of your points differ by 0.4 °C. Look at the two readings from the same place. Is the difference real? How do you know?
2. All the readings were higher on Thursday than on Wednesday. Does that tell you anything about the individual places? What would you do with the numbers to find out?
3. Your circuit took a quarter of an hour. Does the first stop have an unfair advantage over the last? How would you check, without changing the equipment?
4. Someone suggests measuring all eight points in one minute, at a run. What gets better, and what breaks?
5. Your classmate pressed **Save the value** without writing anything in the label field, three times. What is in the table now, and why is that a problem only a week later?
6. If you had to give your data set to another school and ask them to reproduce it, what exactly would you need to write down? Make the list.

7.5 Ideas for assessment along the way

- **Low stakes:** the student receives **the group's chart, already printed**, marks the highest point and the lowest, reads their values off the axis, and writes one sentence for each. No calculation.
- **Standard:** a one-page account — the room plan with the points numbered, the table of values, the chart and a paragraph stating *which* differences are real and which fall within the measurement's spread, with the threshold calculated, not assumed.
- **Extension:** students measure **two quantities** at every point — for example illuminance and temperature —, export a single file, split it by the `quantity` column, join the two quantities by label and look for a link with a scatter plot; or they run the week-long version, normalise every day to its own mean, and argue, from the normalised data, which places are genuinely anomalous. They close with a proposal to the school, with a cost attached: moving a desk, changing a light fitting, a ventilation routine.

Common pitfalls and misconceptions

Pressing the button the instant you arrive. The single most common mistake. A temperature or CO_2 reading taken a few seconds after moving the box describes the *previous* place. The 60 seconds (90 for CO_2) are timed, and are part of what gets written down.

"The points have vanished!" After the **Quantity:** list is changed, the chart shows only the points of the new quantity. The others are in the table, with their quantity, and are exported with it; going back to the old quantity brings them back on the chart too.

Leaving the label field empty. The app writes `#1`, `#2`, `#3` on its own. A week later nobody remembers what they were, and they can no longer be grouped with anything.

Labels written differently each time. `window`, `Window`, `windo` and `by-window` are four categories in a spreadsheet. The printed list, taped to the clipboard, solves this completely — and is the only thing that does.
Holding the box in the hand, pressed to the chest, or breathing over it. Body heat raises the reading by a degree or more, and a breath holds about 40,000 ppm of CO_2. The box is held by its edges, away from the body, with the head turned away.
Direct sun on the box. A sensor in the sun measures its own sun-warmed casing, not the air, and can read several degrees high. Shade the box with your body or the clipboard, and note on the sheet whether the point was sunlit.
Pressing the button twice at the same place, as a "repeat". This is not a repeat: the two values come from the same second and the same position. A genuine repeat means walking the whole circuit a second time.
Closing the window before exporting. The automatic journal in the `logs` folder saves what it can, but without the class's naming rule and outside the shared folder; hunting for it a week later among dozens of files is a lost lesson. What matters is exported before closing.
Reading the height of the points instead of the numbers. The chart's vertical axis does not start at zero, so two points that look far apart may differ by two tenths of a degree. Read the values on the axis and compare them with the threshold.
Comparing two days without recording the weather. A warm, calm, sunny day cannot be compared with a rainy, windy one. Without the one-line note on conditions, half the analysis is guesswork.
Trusting a single reading. A number with no repeat carries no known uncertainty. Repetition is built into the route, not added at the end.

Safety and classroom management

Low overall risk, but the activity moves students through the building, and that changes the risk profile.
Moving around and supervision. The route is fixed in advance and groups do not deviate from it. In corridors, during lessons, walk quietly; ask the colleague's permission before entering their room. Groups are never sent into boiler rooms, onto roofs, into kitchens or any space normally off-limits to students.
Stairs. Nobody walks down stairs looking at a screen. Close the laptop lid or hand it to a classmate between floors.
Cables and equipment left unattended. USB cables and extension leads never cross a walkway; tape them down. A box left running at a fixed point sits in a spot agreed with the building manager, labelled with the class and your name, somewhere it cannot be knocked over, and never on an escape route or blocking a door.
Outdoors. Check the weather. Electronics and laptops do not go out in the rain. On hot, sunny days the school's usual sun rules apply, and in car parks and delivery areas, watch for vehicles. Wipe the boxes down between groups.

Adaptations for pupils with special educational needs

The route and the labels, ready-made. This is the support route and it reaches the same scientific conclusion. The student receives **the quantity already chosen, the order of stops printed and the label list taped to the clipboard**; their task is to wait, press the button and copy the label correctly. The sheet has its rows **numbered in the route's order**, with the label list printed directly above them; for support, **write the labels into their column yourself before copying the sheet**, so the student has nothing to copy: they write only the number on the screen.
An axis already drawn, for two or three points. Give them an axis drawn on paper

and let them plot, by hand, **two or three** of their own points, alongside the app's chart. Representing a number as a position on a graduated axis is the mathematics lesson behind the activity, and it is learned better with three points than with nine.

Three real roles in the group. Roles rotate at every stop and none of them requires moving quickly:

- **timer keeper** — starts the timer on arrival, announces aloud when the 60 seconds are up, and has the right to stop any rushed press of the button; this is the role with the greatest scientific weight in the group, because it decides when a reading counts;
- **label typist** — writes the label *by copying it from the printed list*, letter by letter, and presses Enter; they are the one who, literally, saves the whole week's analysis;
- **mapper** — marks on the room plan where the stop was, notes the height and writes the conditions row.

A student who cannot carry the box can take any of these three roles, and "put it further from the radiator" is itself an experimental decision — whoever makes it is running the measurement.

Support with writing and typing. Spoken answers are accepted instead of written ones, with the teacher or a classmate writing them on the sheet. For students who find typing difficult, **prepare the short labels** (protocol 1 has the shortest) or give them the role of timer keeper.

Language, routine and predictability. Use the same four words throughout the project — label, stop, wait, repeat — and say the same sentence at every stop: *we stand still, we wait a minute, we write the label, we press.* For students who need predictable routine, a fixed route walked at the same time every day is an advantage, not a constraint: name them for it and leave them the same task every day.

If things do not go to plan

You have one box and thirty students. This is actually protocol 1's normal setup: there is one box, and the class walks its own room with it. Project the screen if you can, or seat the class so everyone can see the laptop at every stop. At each point, a different student is the **screen reader** and another is the **label typist**; everyone records in their own table. The eight stops become eight shared tasks this way, not eight minutes of waiting.

You have twenty minutes, not fifty. This is exactly the short route: stay in the room, six points, the labels written on the board beforehand. Drop the second circuit and the export if you must, but **do not drop the written prediction** or the question "how big does a difference have to be to deserve belief?".

You have no projector. Seat the class in a semicircle around the laptop at every stop, close enough to read the big blue number, and name a **screen reader** who says it aloud. The chart can just as well be drawn on the board at the end, from the shared table — and drawing it by hand is, in fact, an extra lesson.

The sensor will not settle. 60 seconds is normal, 90 for CO_2. If the number is still drifting after three minutes, the cause is almost certainly exposure: direct sun, a radiator, a laptop's air vent, a student's hand or a draught. Move the box 20 centimetres, note on the sheet what changed, and repeat the stop.

The box will not connect on a project day. Do not stop the project. Record the day as missing — **a missing day recorded is data; a missing day hidden is a lie** — and carry on the next day. Check meanwhile the cable, the port selected in the app and, if you are on WiFi, the network.

You cannot leave the room at all. Protocols 1 and 2 both run entirely within a single room and lose nothing of their value. Only protocol 3 needs to travel; it can be reduced to three stops — `empty-room` before the first lesson, `occupied-room` at the lesson's midpoint

and `outside` at an open window, with the box held outside for a minute and a half. The school-wide map is lost; the idea is not.
You have no box at all on the day of the lesson. Use the "What the data should look like" table from this guide as a printed data set: students receive eight labelled values, calculate the spread, the drift and the threshold, and decide which differences are real. **The whole reasoning stays the same**; only their own measurement is lost — and this is the activity in the collection best suited to that loss, since its lesson is about what you do with the numbers, not where they came from.

8. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for the network of stations and observation practice; a good model for why a class needs a fixed protocol sheet.
- **NOAA** — `https://www.noaa.gov` — search for observation-network standards and instrument exposure; explains why the time of day and a sensor's siting matter so much.
- **Institute of Physics** — `https://www.iop.org` — search for practical measurement and uncertainty; the repetition argument in this guide is developed there at length.
- **STEM Learning** — `https://www.stem.org.uk` — search for fieldwork and data-processing resources, suited to a schoolyard study.
- **Copernicus** — `https://www.copernicus.eu` — European environmental data, to place a schoolyard study on the scale of a city or a continent.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for simulations on representing and interpreting data, useful before the first lesson of analysis in the spreadsheet.

9. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Because the activity spans two levels, the descriptors read as follows:

- **Measurement practice** — the student waits the timed 60 seconds, holds the box by its edges and in the shade, writes the label exactly from the list, measures the height instead of guessing it, and records the conditions alongside the number. At Scientist level, in addition: exports and names the file according to the class's rule, on the day itself.
- **Data interpretation** — at Explorer, the student reads the chart, says which value is the biggest and which the smallest, reading the numbers on the axis, and knows that two identical labels are two points, not one. At Scientist: calculates the repeat spread and the drift, formulates the decision threshold, and normalises to the day's mean before comparing two days.
- **Scientific reasoning** — the student distinguishes *"we could not detect a difference"* from *"there is no difference"*; explains why a reading taken right after moving describes the previous place; and says what would need to be written down for someone else to reproduce the

measurement.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

10. The student sheet

The sheet the students fill in is a separate document: `docs/en/student_sheets/measurement_notebook.pdf`. It prints double-sided, one sheet of paper per student, and is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — asking for the prediction **before** the first reading.

Four things worth knowing before printing it. **The six label-writing rules sit on the sheet, not only in the guide**, right above the table, because that is where the student looks at the moment of typing. **The label lists for the three protocols are printed on the sheet, word for word and numbered**, in the order walked; the student *circles* the chosen protocol and copies from there, letter by letter — so there are no labels invented on the spot. **The table's rows are numbered in the same order**, so row n is stop n from the circled list, and the table has **two columns of values**, one for each circuit, precisely because repetition needs to exist in the sheet's form, not in the student's goodwill. And **the threshold calculation has a worked example right beside the blank row**: the spread plus the drift, with numbers, so the student sees once what it looks like. The tables are not repeated here precisely so that the two documents cannot drift apart: there is a single printable version, and it is the sheet.